

Lucas Torres Santana

Head of Staking at Hashdex

Rio de Janeiro, Brazil
linkedin.com/in/lucas-santana-90536295
github.com/lucastsantana
lucastsantana.com

SUMMARY

Bridges the gap between complex theoretical frameworks and actionable digital asset strategies, backed by over four years of deep industry experience in crypto research and asset management, complemented by two years of specialized research and modeling within the oil and gas industry. A rigorous scientific thinker holding both a BSc and a PhD in Theoretical Physics from the Federal University of Rio de Janeiro, applying a first-principles approach to problem-solving and extracting meaningful insights from complex data across diverse fields, including digital assets, cosmology, and geosciences. Thrives at the intersection of technical depth and clear communication, delivering high-stakes inputs for executive leadership and global clients, making the complex accessible.

EXPERIENCE

Hashdex	Rio de Janeiro, Brazil
Head of Staking	Dec 2025 – Present
Lead staking infrastructure strategy and operations across Hashdex's crypto index products. Oversee validator node management, staking yield optimization, and protocol research across major Proof-of-Stake networks. Drive product development for regulated crypto ETFs available in Brazil and internationally.	
Research Lead	Dec 2024 – Nov 2025
Led quantitative and qualitative research on Bitcoin, Ethereum, Layer-1 alternatives, DeFi, and NFTs to support product and investment decisions. Delivered high-stakes research inputs for executive leadership and global clients.	
Research Analyst	Sep 2022 – Nov 2024
Conducted extensive market research on Bitcoin, Ethereum, alternative Layer-1s (EVM and non-EVM), scaling solutions, DeFi, and NFTs. Provided in-depth technical knowledge for staking operations (bonding/unbonding periods, entry/exit churns, sweeping cycles). Used Python and SQL to retrieve and process on-chain and market data from Dune Analytics, Glassnode, and CryptoQuant.	
Junior Research Analyst II	Dec 2021 – Aug 2022
Wrote a Primer on Smart Contracts and helped create marketing materials for the launch of a crypto ETF in Brazil. Participated in investor calls showcasing the investment thesis for Layer-1s. Built workflows to gather on-chain and crypto market data, and provided regular updates to internal stakeholders and external clients.	
CGG	Rio de Janeiro, Brazil · Houston, TX, United States
Geophysicist II	Dec 2020 – Sep 2021
Worked on production and research projects for datasets with different acquisition types, handling preprocessing and velocity model building routines including full-waveform inversion, ray-based tomography, and interpretation. Daily workflow involved Python, shell scripting, and simulations in HPC environments.	
Geophysicist I	Apr 2020 – Nov 2020
Developed analytical workflows and data intelligence pipelines for a new imaging technology under evaluation at the Rio center.	
Geophysicist I	Oct 2019 – Mar 2020
Gained hands-on experience in seismic processing and velocity model building, contributing to a production project at the Houston center targeting the Gulf of Mexico.	

EDUCATION

PhD in Theoretical and Mathematical Physics

2017 – 2023

Federal University of Rio de Janeiro (UFRJ)

Thesis: *"Contributions to the description of linear waves in curved spacetimes"*

Advisors: Prof. Dr. Ribamar R. R. dos Reis, Prof. Dr. Maurício O. Calvão

Bachelor of Science in Physics

2013 – 2016

Federal University of Rio de Janeiro (UFRJ)

Magna Cum Laude

GPA 9.1/10.0

SELECTED PUBLICATIONS

Unified Approach to Coupled Homogeneous Linear Wave Propagation in Generic Gravity (1 citations)

João C. Lobato, Lucas T. Santana, Maurício O. Calvão, Ribamar R.R. Reis — *Physical Review D*, 110, 044031 (2024) · DOI: 10.1103/PhysRevD.110.044031

Influence of Gravitational Waves Upon Light in the Minkowski Background (4 citations)

João C. Lobato, Isabela S. Matos, Lucas T. Santana, Ribamar R.R. Reis, Maurício O. Calvão — *Physical Review D*, 104, 024024 (2021) · DOI: 10.1103/PhysRevD.104.024024

Evolution of the Electric Field Along Null Rays (7 citations)

João C. Lobato, Isabela S. Matos, Lucas T. Santana, Maurício O. Calvão, Ribamar R.R. Reis — *Physical Review D*, 101, 081501 (2020) · DOI: 10.1103/PhysRevD.101.081501

How Does Light Move in a Generic Metric-Affine Background? (24 citations)

Lucas T. Santana, Maurício O. Calvão, Ribamar R.R. Reis, Beatriz B. Siffert — *Physical Review D*, 95, 061501 (2017) · DOI: 10.1103/PhysRevD.95.061501

SKILLS & LANGUAGES

TECHNICAL

Python

SQL

Dune Analytics

Glassnode

CryptoQuant

Plotly

Shell scripting

HPC

DOMAIN

Staking

DeFi

ETFs

On-chain Analytics

Seismic Processing

Full-Waveform Inversion

General Relativity

LANGUAGES

Portuguese · Native

English · Native / Bilingual

French · Professional Working

Spanish · Limited Working